



AI INNOVATION HACKATHON 2026

AI/ML Methodology

STANDARDIZED PROBLEM STATEMENTS

Women in AI, Research, Innovation & Entrepreneurship Club

Research • Innovation • Entrepreneurship

5 Challenges • Online • AI Innovation Hackathon 2026

Each challenge follows the same structured format for clarity, consistency and ease of review.

CHALLENGE PORTFOLIO

NO.	CHALLENGE	DIFFICULTY
1	CrisisSense AI — Multimodal Disaster Impact Forecasting	Advanced
2	HeatShield AI — Hyperlocal Urban Heat Risk Forecasting	Intermediate to Advanced
3	SupplyShield AI — Essential Supply Disruption Prediction	Advanced
4	CyberTrace AI — Behavioral Threat Detection Without Signature Dependence	Intermediate to Advanced
5	ResourceMind AI — Adaptive Allocation Under Changing Demand	Advanced



CHALLENGE 01

CrisisSense AI — Multimodal Disaster Impact Forecasting

DOMAIN

Disaster / Geospatial AI

DIFFICULTY

Advanced

PROBLEM STATEMENT

During floods, landslides and severe storms, decision-makers often receive fragmented information from weather feeds, satellite imagery, sensors and public reports. The challenge is to estimate where disruption is likely to become severe before ground conditions fully reveal the impact. A useful system must handle incomplete, noisy and rapidly changing information rather than relying on one source alone.

EXPECTED SOLUTION

Participants should build a multimodal AI system that combines heterogeneous signals to forecast high-risk zones, estimate likely impact severity and highlight the evidence behind each prediction. The system should continuously update its assessment as new observations become available.

SUGGESTED DATASETS / RESOURCES

Public satellite imagery; weather/open-data feeds; disaster datasets; terrain/elevation data; documented synthetic sensor streams; controlled synthetic scenarios for missing-data and changing-condition cases.

AI / ML COMPONENT

Multimodal learning; geospatial modelling; time-series forecasting; anomaly detection; uncertainty estimation; explainable AI.

WHAT MAKES THE CHALLENGE INTERESTING?

Moves beyond detecting a disaster after it happens by combining different data modalities to reason about evolving impact, uncertainty and spatial differences in risk.

SUGGESTED ON-SCREEN DEMONSTRATION

Load/update weather, imagery and sensor inputs map high-risk zones show impact severity display contributing signals update prediction when an input changes show uncertainty.

TECHNICAL REQUIREMENTS

Python; geospatial and raster/vector processing; multimodal data ingestion; model training/inference; visualization/map interface; reproducible preprocessing; uncertainty reporting.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Ingest at least two data modalities and produce a spatial risk/impact forecast.
- Expose key contributing signals and communicate uncertainty clearly.

OPTIONAL ADVANCED FEATURES

- Test missing-data scenarios and changing-condition scenarios.
- Add temporal forecasting and scenario simulation.
- Compare single-modality and multimodal performance.
- Add uncertainty calibration or richer explanation views.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Responsible AI / Ethical Considerations (proposed supporting detail): Use only lawful and permitted datasets; document dataset limitations and bias; protect personal or sensitive information; do not make unsupported real-world claims; clearly communicate uncertainty; and distinguish a hackathon research/educational prototype from a production decision system.

INNOVATION ANGLE

Multimodal, continuously updated risk forecasting with explicit uncertainty and evidence.

SUCCESS / EVALUATION METRICS

Forecast quality, spatial error/overlap where applicable, calibration/uncertainty quality, update responsiveness, and robustness to missing/noisy inputs.



CHALLENGE 02

HeatShield AI — Hyperlocal Urban Heat Risk Forecasting

DOMAIN

Urban Climate / Geospatial AI

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

Urban heat is not distributed evenly. Buildings, roads, vegetation, traffic and local weather can create sharp differences in thermal exposure across nearby areas, making city-wide averages insufficient for targeted planning. The challenge is to identify and forecast these local variations.

EXPECTED SOLUTION

Participants should develop an AI system that predicts hyperlocal heat-risk levels using environmental and urban features, identifies hotspots and forecasts how risk may change over time. The system should allow users to inspect why a particular area receives a higher risk level.

SUGGESTED DATASETS / RESOURCES

Satellite land-surface-temperature data; weather data; OpenStreetMap/urban features; vegetation indices; permitted public infrastructure datasets; historical observations or documented synthetic data.

AI / ML COMPONENT

Geospatial machine learning; spatial-temporal forecasting; feature fusion; clustering; interpretable risk modelling.

WHAT MAKES THE CHALLENGE INTERESTING?

Creates a fine-grained decision layer for urban heat rather than relying only on broad weather forecasts or static heat maps, allowing the model to learn relationships between built environment and thermal risk.

SUGGESTED ON-SCREEN DEMONSTRATION

Select area/time map predicted heat-risk levels identify hotspots show contributing factors change a scenario such as vegetation/built-up/weather display updated risk.

TECHNICAL REQUIREMENTS

Python; geospatial processing; spatial-temporal modelling; map visualization; feature engineering; scenario-analysis interface.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Generate a map-based heat-risk forecast and identify hotspots.
- Explain major contributing factors and support time-based prediction.

OPTIONAL ADVANCED FEATURES

- Add scenario simulation for vegetation, built-up area or weather changes.
- Compare spatial and temporal modelling approaches.
- Provide uncertainty or confidence maps.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

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INNOVATION ANGLE

Hyperlocal, scenario-aware heat-risk forecasting rather than static or city-wide heat maps.

SUCCESS / EVALUATION METRICS

Forecast error, hotspot detection quality, temporal prediction quality, explanation usefulness, and scenario consistency.



CHALLENGE 03

SupplyShield AI — Essential Supply Disruption Prediction

DOMAIN

Supply Chain / Predictive AI

DIFFICULTY

Advanced

PROBLEM STATEMENT

Shortages of essential goods can emerge from delayed deliveries, unusual demand, weather events, transport disruption or cascading failures. Organizations often react after inventory levels have already become critical. The challenge is to identify warning patterns before a shortage becomes severe.

EXPECTED SOLUTION

Participants should build a predictive system that learns demand, supply and disruption patterns to estimate the probability and timing of future shortages and identify the most influential factors. The system should connect related events rather than treating every shortage as an isolated incident.

SUGGESTED DATASETS / RESOURCES

Public supply-chain datasets; retail demand datasets; transportation/open-data sources; weather/event data; documented synthetic supply networks.

AI / ML COMPONENT

Time-series forecasting; graph-based modelling; anomaly detection; probabilistic prediction; explainable feature attribution.

WHAT MAKES THE CHALLENGE INTERESTING?

Treats shortages as a connected-system prediction problem rather than a simple inventory-threshold problem, allowing cascading effects and changing demand to be represented.

SUGGESTED ON-SCREEN DEMONSTRATION

Select item/location forecast disruption probability and time-to-shortage visualize related events/network factors show contributing factors and uncertainty.

TECHNICAL REQUIREMENTS

Python; time-series/graph data processing; forecasting; event correlation; visualization; uncertainty/confidence reporting.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Forecast disruption risk for selected items or locations.
- Estimate expected time to critical shortage and show contributing factors with confidence/uncertainty.

OPTIONAL ADVANCED FEATURES

- Model cascading disruptions through a graph.
- Add probabilistic forecasts and scenario simulation.
- Compare isolated-item modelling with connected-system modelling.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

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INNOVATION ANGLE

Connected-system forecasting that models cascading effects and changing demand.

SUCCESS / EVALUATION METRICS

Forecast accuracy, time-to-event error, disruption-risk calibration, detection of early-warning signals, and robustness to missing events.



CHALLENGE 04

CyberTrace AI — Behavioral Threat Detection Without Signature Dependence

DOMAIN

Cybersecurity / Behavioral Detection

DIFFICULTY

Intermediate to Advanced

PROBLEM STATEMENT

Traditional security systems can struggle with previously unseen attacks because they often depend on known signatures or fixed rules. Abnormal behavior may appear as subtle changes across users, devices, processes or network activity and may only become meaningful when several events are considered together.

EXPECTED SOLUTION

Participants should develop a behavior-based detection engine that learns normal activity patterns, identifies suspicious deviations and correlates related events into meaningful security incidents. The system should distinguish isolated anomalies from patterns that deserve investigation.

SUGGESTED DATASETS / RESOURCES

Permitted cybersecurity intrusion datasets; system/network logs; authentication-event datasets; synthetic enterprise activity generated for the challenge.

AI / ML COMPONENT

Unsupervised anomaly detection; sequence modelling; graph analytics; event correlation; representation learning; explainable detection.

WHAT MAKES THE CHALLENGE INTERESTING?

Focuses on behavioral evidence and cross-event correlation instead of relying only on predefined attack signatures, making the challenge suitable for unknown or evolving patterns.

SUGGESTED ON-SCREEN DEMONSTRATION

Upload/stream logs learn/display baseline detect anomalies correlate events into an incident assign risk show timeline and triggering behavioral signals.

TECHNICAL REQUIREMENTS

Python; log/event parsing; anomaly detection; sequence/graph processing; event correlation; security-safe sandboxed evaluation; dashboard.



SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Process event streams or logs and group related anomalies into incidents.
- Provide a risk level, event timeline and explanation of behavioral signals that triggered the alert.

OPTIONAL ADVANCED FEATURES

- Compare multiple anomaly-detection approaches.
- Add graph-based incident correlation.
- Test against synthetic previously unseen behavioral patterns.
- Add analyst feedback or threshold tuning.

RESPONSIBLE AI / ETHICAL CONSIDERATIONS

Use only permitted datasets and controlled/synthetic environments. Do not test against systems without authorization. Do not include real credentials, secrets, personal data or live attack activity. Document false positives and limitations.

INNOVATION ANGLE

Behavioral, cross-event detection that does not depend solely on known signatures.

SUCCESS / EVALUATION METRICS

Detection precision/recall or appropriate anomaly metrics, false-positive rate, incident grouping quality, detection latency, and explanation quality.



CHALLENGE 05

ResourceMind AI — Adaptive Allocation Under Changing Demand

DOMAIN

Resource Allocation / Decision Intelligence

DIFFICULTY

Advanced

PROBLEM STATEMENT

Public and organizational resources such as emergency teams, charging stations, shelters, transport capacity or maintenance crews may be distributed inefficiently when demand changes rapidly. A decision made using yesterday's demand may become unsuitable when conditions change.

EXPECTED SOLUTION

Participants should create an AI-assisted allocation system that predicts near-future demand, identifies resource gaps and recommends allocation plans while considering constraints and changing conditions. The system should allow different scenarios to be simulated before an allocation decision is made.

SUGGESTED DATASETS / RESOURCES

Public mobility/demand datasets; service-request datasets; geographic data; historical utilization records; synthetic scenarios for resource constraints.

AI / ML COMPONENT

Demand forecasting; optimization; reinforcement learning or constrained decision modelling; geospatial analytics; scenario simulation.

WHAT MAKES THE CHALLENGE INTERESTING?

Combines prediction with constrained decision-making, requiring the system to reason about what action should be taken after forecasting demand instead of stopping at prediction.

SUGGESTED ON-SCREEN DEMONSTRATION

Select scenario forecast demand display resource gaps generate allocation plan show constraints/trade-offs simulate an alternative scenario.

TECHNICAL REQUIREMENTS

Python; forecasting; optimization/decision modelling; constraint handling; geospatial visualization; scenario simulator.

SUGGESTED MINIMUM DELIVERABLE

- Working prototype with reproducible source code/notebook.
- README describing the problem, data, methodology, implementation and limitations.
- Demonstration of the core AI/ML functionality on representative inputs.
- Evaluation results using appropriate metrics for the selected task.
- Forecast demand and simulate alternative allocation strategies.
- Show constraint violations/trade-offs and provide an interpretable recommended plan.

OPTIONAL ADVANCED FEATURES

- Compare optimization and learning-based allocation.
- Add multi-period planning.
- Introduce changing constraints and scenario stress tests.
- Provide sensitivity analysis for resource availability.



RESPONSIBLE AI / ETHICAL CONSIDERATIONS

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INNOVATION ANGLE

Links prediction directly to constrained, explainable action planning.

SUCCESS / EVALUATION METRICS

Forecast quality, constraint satisfaction, allocation efficiency, service coverage, robustness under changing demand, and computational cost.